

Credit History, Standardization, and Racial Disparities in Mortgage Approval: Evidence on Underwriting Behavior, Loan-Level Heterogeneity, and Credit Allocation Efficiency

Author Identity Redacted for Peer Review

Abstract

We examine the locus and structure of racial disparities in U.S. mortgage approvals using 2018 to 2023 Home Mortgage Disclosure Act records from ten large depository institutions, with a focused study of Truist Bank’s 2021 to 2023 portfolio (336,332 applications, of which 32,703 are from Black applicants). After controlling for income, loan amount, debt-to-income ratio, loan-to-value ratio, loan type, occupancy, automated underwriting system, conforming-loan status, and missingness indicators, Black applicants face an adjusted approval odds ratio of 0.575 (95% CI [0.555, 0.595], $p < 0.001$) relative to comparable White applicants at Truist. The disparity is stable across geographic specifications, with state fixed effects yielding OR = 0.520 and within-county linear probability models yielding a coefficient of -0.114 percentage points. Decomposing the denial process via a binary logit framework with originated applicants as the reference category, Black applicants are 3.17 times more likely to receive a credit history denial and 1.69 times more likely to receive a debt-to-income denial; the probability of a collateral-based denial is statistically indistinguishable from White applicants ($p = 0.363$). The disparity is monotonically larger in smaller loans, ranging from OR = 0.491 in the bottom loan-size quartile (median \$45,000) to OR = 0.748 in the top quartile (median \$435,000). A formal interaction confirms a 19% amplification of the racial penalty in the small-loan segment ($p < 0.001$). Cross-institution variation is substantial: institution-level Black-White approval gaps range from -5.7 percentage points (Bank of America) to -17.9 percentage points (PNC Bank), with a Pearson correlation

of $r = 0.746$ between log institution size and gap magnitude. An automated underwriting analysis shows manual or unspecified-AUS underwriting yields a Black OR of 0.524 versus 0.701 for Loan Prospector-routed applications, consistent with a standardization mechanism. We benchmark the small-loan rate premium against the published Fannie Mae Loan-Level Price Adjustment schedule and find that LLPA accounts for only 1.4% (0.14 of 10.10 basis points) of the observed premium, weakening the mechanical-pricing interpretation. Government-sponsored enterprise performance data show small loans carry a statistically significant but economically small 60-day delinquency premium of 0.066 percentage points after FICO/DTI/LTV controls (base rate 0.31%), alongside materially higher FICO interquartile-range dispersion in the small-loan segment (61 versus 48 points). We conclude with a counterfactual quantification: applying the expanded-controls model, Truist’s racial approval gap implies approximately 1,293 excess Black denials annually and \$148M to \$331M in foregone wealth accumulation under a range of home-equity scenarios. We discuss three remaining identification challenges (credit-score-conditional mediation, loan-level profitability parity, and credit-file-depth heterogeneity) that require data not in the public HMDA record and identify the FHFA National Mortgage Database as the natural next-step data source.

Keywords: fair lending, mortgage approval, racial disparities, HMDA, automated underwriting, credit allocation

JEL Classification: G21, G28, J15, R31

1 Introduction

The Equal Credit Opportunity Act of 1974 and the Fair Housing Act of 1968 prohibit discrimination in mortgage lending on the basis of race, ethnicity, color, religion, national origin, sex, marital status, or age. Five decades after these statutes, raw approval-rate disparities by race persist at large U.S. depository institutions, and a recent literature documents that risk-adjusted disparities also remain substantial (Bartlett et al., 2022; Bhutta et al., 2022). Two questions remain open in this literature. First, where in the underwriting process do racial disparities arise: in front-end automated screening, in manual review of credit history, in collateral assessment, or in pricing? Second, how do these disparities vary across loan sizes and across institutions, and what mechanisms can be ruled in or out using publicly available regulatory data?

This paper addresses both questions using HMDA records covering ten large U.S. depository institutions over 2018 to 2023 (10.26 million applications after restricting to originated and denied actions), supplemented with Fannie Mae loan-level acquisition and performance data covering 72.7 million loan-month observations from the 2021Q1 acquisition cohort. The HMDA records, expanded substantially under the 2018 reporting amendments, now include loan-to-value ratio, automated underwriting system identifier, loan type, occupancy, lien status, loan term, and rate spread, all of which were unavailable in pre-2018 fair lending studies that relied on income, loan amount, debt-to-income, and loan purpose alone. We exploit this expanded covariate set to address omitted-variable concerns that have limited the interpretability of HMDA-based fair lending estimates.

Our principal findings are as follows. First, after expanding the control set to include LTV, loan type (FHA, VA, USDA, conventional), occupancy, manufactured housing, conforming-loan status, AUS, and missingness indicators, the Black-White adjusted approval odds ratio at Truist Bank is 0.575 (95% CI [0.555, 0.595], $p < 0.001$). An alternative version of the same model estimated with listwise deletion of incomplete records yields a comparable Black OR of 0.526, indicating that the headline estimate is not an artifact of how missingness is handled. The disparity is robust to state fixed effects (OR = 0.520), within-county demeaning (LPM coefficient -0.114 pp), and exclusion of FHA, VA, USDA, and manufactured housing loans (OR = 0.565). Listwise deletion is not a concern: missingness rates for income, loan amount, DTI, and LTV are below 0.1% in the

post-2018 sample for all racial groups. We therefore reject the omitted-variable explanation as the primary driver of the residual disparity.

Second, we decompose the denial process by estimating a separate binary logit for each denial reason using originated applications as the reference category. This specification is motivated by the concern that conditioning on denial introduces selection bias. Black applicants are 3.17 times more likely than comparable White applicants to receive a credit history denial relative to origination (95% CI [3.07, 3.29]), 1.69 times more likely to receive a DTI denial, and 1.40 times more likely to receive an incomplete-application denial. The probability of a collateral-based denial is statistically indistinguishable between Black and White applicants ($OR = 1.04$, $p = 0.363$), ruling out appraisal bias as a meaningful contributor at this institution and pointing to credit history evaluation as the dominant locus of disparity.

Third, the Black-White approval gap is monotonically larger in smaller loans. Quartile-stratified estimates yield Black $OR = 0.491$ in the smallest quartile (median loan \$45,000), rising to 0.748 in the largest quartile (median \$435,000). A formal $Black \times LoanQ1$ interaction term is statistically significant ($OR = 0.807$, $p < 0.001$), implying a 19% amplification of the racial penalty in small loans. Cross-institution variation is similarly substantial: pooled estimates with institution and year fixed effects yield institution-specific Black coefficients ranging from -5.7 pp at Bank of America to -17.9 pp at PNC Bank, with a Pearson correlation of 0.746 between log institution size and gap magnitude. An automated-underwriting decomposition shows Loan Prospector-routed applications produce Black $OR = 0.701$ versus 0.524 for manual or unspecified-AUS routing, consistent with the standardization-reduces-disparity hypothesis.

Fourth, we benchmark the small-loan rate premium documented in our 2021Q1 GSE sample (10.10 basis points) against Fannie Mae’s published 2022 LLPA schedule and find that the standard 25 basis point small-balance LLPA, when applied at the loan level, accounts for only 0.14 basis points (1.4%) of the observed premium. The residual 9.96 basis points survives FICO/DTI/LTV controls. This narrows the space of explanations for the rate premium: it is not predominantly a mechanical pass-through of GSE pricing rules. Outcome-based evidence is mixed. We document a statistically significant 0.066 percentage point 60-day delinquency premium for small loans after FICO/DTI/LTV controls, which is real but economically small relative to a 0.31% base rate.

We additionally document materially higher FICO interquartile-range dispersion in the small-loan quartile ($\text{IQR} = 61$ points) than in the largest quartile ($\text{IQR} = 48$ points).

Fifth, we quantify the wealth implications. Applying the expanded-controls counterfactual to Truist’s 2021 to 2023 portfolio implies approximately 3,878 excess Black denials over three years, or 1,293 annually. Translated to home-equity accumulation using the FHFA all-transactions index (Method A) and a full equity-build model (Method B), the annual wealth foregone ranges from \$148.3M to \$330.8M.

Three identification challenges remain that we cannot address with public data. We cannot conduct a causal mediation decomposition that separates direct discriminatory taste from indirect channels operating through credit scores, because HMDA does not include credit scores. We cannot conduct a true loan-level profitability parity test, because matched origination, performance, loss-given-default, and servicing data with race-of-applicant identifiers are not publicly available. And we cannot directly measure credit-file depth or thin-file status, which would tighten the variance-based mechanism. We discuss these gaps in Section 7 and identify the FHFA National Mortgage Database as the natural next-step data source.

The paper proceeds as follows. Section 2 situates our contribution. Section 3 describes data sources, sample construction, and variable definitions. Section 4 sets out the empirical specifications. Section 5 reports baseline, expanded, and robustness estimates including the denial-decomposition, geographic-FE, AUS, cross-institution, LLPA, and delinquency analyses. Section 6 translates the statistical disparities into wealth-accumulation magnitudes. Section 7 discusses limitations and unresolved identification questions. Section 8 concludes.

2 Related Work

Bartlett et al. (2022) use HMDA combined with proprietary lender data to show that algorithmic lenders narrow but do not eliminate racial pricing disparities. Bhutta et al. (2022) document persistent risk-adjusted denial gaps using HMDA-based methods. Munnell et al. (1996) established the foundational result that minority applicants face elevated denial probabilities after observable risk controls in the Boston Fed data. Our contribution differs along three dimensions. First, we

exploit the post-2018 HMDA expansion to incorporate LTV, AUS, occupancy, lien status, and loan type, which were unavailable in earlier studies and which the literature has identified as plausible omitted variables. Second, we directly test the standardization-reduces-disparity hypothesis by exploiting within-bank variation in AUS routing and across-bank variation in size. Third, we benchmark the small-loan rate premium against the published GSE LLPA schedule, an analysis that to our knowledge has not previously appeared in the fair-lending literature.

The mechanism by which thin-file applicants face stricter approval thresholds, in turn producing both higher denial rates and pricing premia even at equal medians of observable risk metrics, has received attention in the credit-allocation literature. Our finding that FICO interquartile range is 27% wider in the smallest loan quartile than in the largest provides indirect evidence consistent with this mechanism, though we cannot directly observe trade-line counts or file depth in HMDA or in the GSE acquisition data.

A growing literature emphasizes outcome-based discrimination tests as a complement to approval-rate parity. We substitute a delinquency-based outcome test in Section 5.10 as a partial approximation to a full profitability parity test, the latter of which is infeasible without loan-level loss data.

3 Data

3.1 HMDA Application Records

We use HMDA Loan Application Register data from 2018 through 2023, downloaded via the Consumer Financial Protection Bureau HMDA Data Browser. The 2018 reporting amendments substantially expanded the dataset, adding loan-to-value ratio, AUS identifier, loan type, occupancy, manufactured housing flag, lien status, loan term, conforming-loan limit indicator, rate spread, and other fields not previously reported. We restrict the universe to action codes 1 (originated) and 3 (denied), excluding withdrawn applications, applications closed for incompleteness, and approved-but-not-accepted offers, in order to focus on lender decisions rather than applicant-driven outcomes.

The combined 2018 to 2023 panel covers ten large depository institutions: Bank of America,

Citizens Bank, Fifth Third Bank, Huntington Bank, JPMorgan Chase, PNC Bank, Regions Bank, Truist Bank (including pre-merger BB&T and SunTrust LEIs), U.S. Bank, and Wells Fargo. Total observations after the action-code filter and institution restriction: 10,261,064 applications. Our focused Truist sample covers 2021 to 2023, the post-merger reporting period, and contains 336,332 applications, of which 32,703 are from Black applicants.

3.2 Race and Ethnicity Construction

We use the HMDA-derived race and ethnicity fields, which apply the agency’s standard hierarchy to applicant and co-applicant self-reports. Following the standard convention, an application is classified as Black if the primary applicant’s derived race is Black or African American, as Hispanic if the derived ethnicity contains Hispanic, and as Asian if the derived race is Asian. White (non-Hispanic) is the reference category. Applicants whose race or ethnicity is not provided or not applicable are retained in the sample with their indicator variables set to zero, which conservatively biases against finding a racial effect.

3.3 Variable Definitions

Outcome. *Approved* equals 1 for action code 1 and 0 for action code 3. We additionally analyze the categorical denial reason field (*denial_reason-1*), which records the primary HMDA-coded denial reason: 1 = debt-to-income, 2 = employment history, 3 = credit history, 4 = collateral, 5 = insufficient cash, 6 = unverifiable information, 7 = credit application incomplete, 8 = mortgage insurance denied, 9 = other, 10 = not applicable.

Credit and underwriting controls. Log income, log loan amount, DTI midpoint (computed by parsing the HMDA categorical DTI ranges and taking the bin midpoint), LTV, loan purpose dummies (purchase, refinance, cash-out refinance), loan type dummies (FHA, VA, USDA), occupancy dummies (investment, second home, with primary residence as the omitted category), manufactured housing flag, conforming-loan-limit flag, lien status (subordinate vs first), 30-year fixed loan term flag, and AUS dummies (Desktop Underwriter, Loan Prospector, manual or unspecified). We additionally include indicator variables for missingness in LTV, DTI, and income to retain observations with missing values; missingness rates for these variables are below 0.1% in the

post-2018 sample for all racial groups, eliminating selection-into-completeness as a concern.

3.4 GSE Performance Data

We supplement HMDA with Fannie Mae loan-level acquisition and performance files for 2021Q1, downloaded from the Fannie Mae Single-Family Historical Loan Performance dataset (Fannie Mae, 2024). The acquisition file contains origination characteristics (FICO, original LTV, original UPB, original DTI, loan term, loan purpose, occupancy, property state, seller name) for each loan acquired by Fannie Mae in that quarter. The performance file tracks delinquency status by month for each loan; we use the cumulative 60-day delinquency indicator as our primary outcome. The raw 2021Q1 performance file contains 73,467,360 loan-month records; after filtering to records with valid FICO, original UPB, and origination period, the analysis sample retains 72,747,007 loan-month observations.

4 Methods

4.1 Baseline Logistic Regression

The primary specification is

$$\Pr(\text{approved}_i = 1) = \Lambda(\alpha + \mathbf{X}_i' \boldsymbol{\beta} + \mathbf{R}_i' \boldsymbol{\gamma}),$$

where $\Lambda(\cdot)$ is the logistic CDF, $\mathbf{X}_i$ is the credit-and-underwriting control vector defined in Section 3, and $\mathbf{R}_i$ is the vector of race/ethnicity indicators. The coefficient of primary interest is the Black indicator, exponentiated to an odds ratio $\text{OR}_{\text{Black}} = \exp(\gamma_{\text{Black}})$. Standard errors are computed via the inverse Hessian of the log-likelihood; we report HC3-robust variants in the linear-probability robustness specifications.

4.2 Geographic Robustness

To address the concern that geographic composition (Truist’s heavy presence in southeastern markets with distinct demographic profiles) drives the residual estimate, we estimate three nested

specifications. The baseline includes no geographic controls. The state-fixed-effects specification adds state dummies absorbing all state-level unobserved heterogeneity. The county-fixed-effects specification, due to the memory burden of estimating logits with several hundred dummies on the Truist sample, is implemented as a within-county-demeaned linear probability model: each variable is demeaned by county, and an OLS regression with HC3-robust standard errors recovers the within-county effect of race on approval probability. The LPM coefficient is interpreted as the percentage-point change in approval probability rather than as an odds ratio.

4.3 Denial Reason Decomposition

The original denial-conditional analysis estimates the relative probability of a credit history versus DTI citation among denied applicants. This conditions on the outcome of the approval decision, which is itself a function of race, and therefore introduces selection bias. We replace this with a series of binary logits in which the outcome is 1 if the application received the specified denial reason and 0 if originated. Each model is estimated on the subsample of applications that were either originated or received the specific denial reason in question, with full credit-and-underwriting controls. This specification directly tests whether, conditional on financial characteristics, Black applicants are differentially likely to receive each denial reason relative to being originated.

4.4 AUS Stratification

We stratify the sample by primary AUS (Desktop Underwriter, Loan Prospector, manual or unspecified) and re-estimate the baseline expanded-controls logit within each stratum. The Black coefficient under each AUS routing tests whether automated underwriting systems compress disparities relative to manual review.

4.5 Cross-Institution Pooled Model

We estimate a pooled linear probability model on all ten institutions over 2018 to 2023:

$$\text{approved}_{ij} = \alpha + \mathbf{X}'_{ij}\boldsymbol{\beta} + \mathbf{R}'_{ij}\boldsymbol{\gamma} + \mu_j + \tau_t + \varepsilon_{ij},$$

where μ_j is an institution fixed effect, τ_t is a year fixed effect, and standard errors are HC3-robust. We additionally estimate the same specification separately within each institution to recover institution-specific Black coefficients, and we report the Pearson correlation between log institution size and gap magnitude as a summary measure of the size-standardization relationship.

4.6 Event Study

To examine the BB&T-SunTrust merger as a potential shock to underwriting standardization, we estimate

$$\text{approved}_{ijt} = \alpha + \sum_{t \neq 2018} \delta_t \cdot (\text{Black}_i \times \text{Truist}_j \times \mathbf{1}[t]) + \mathbf{X}'_{ijt}\boldsymbol{\beta} + \mathbf{R}'_{ijt}\boldsymbol{\gamma} + \mu_j + \tau_t + \varepsilon_{ijt}$$

on the pooled 2018 to 2023 panel covering Truist plus six control institutions. Standard errors are clustered at the institution level. The coefficients δ_t trace the dynamic evolution of the Black-White approval gap at Truist relative to controls, with 2018 as the omitted reference. We test pre-trend parallelism by examining whether δ_{2019} and δ_{2020} are individually statistically distinguishable from zero.

4.7 LLPA Benchmark

We construct a loan-by-loan LLPA estimate for the 2021Q1 GSE sample using Fannie Mae’s published April 2022 LLPA matrix (Fannie Mae, 2022), which assigns a price adjustment in basis points to each FICO-by-LTV cell, plus a flat 25 basis point small-balance LLPA for loans below \$150,000. We then estimate

$$\text{rate}_i = \alpha + \beta_1 \text{FICO}_i + \beta_2 \text{DTI}_i + \beta_3 \text{LTV}_i + \beta_4 \log \text{UPB}_i + \beta_5 \text{Small}_i + (\beta_6 \text{LLPA}_i) + \varepsilon_i$$

both with and without the LLPA control. The coefficient β_5 on the small-loan indicator captures the residual rate premium not explained by the included risk and pricing controls. The reduction in β_5 when LLPA is added to the model quantifies the share of the small-loan rate premium that is attributable to the published GSE pricing schedule.

4.8 Delinquency Analysis

For the 2021Q1 GSE acquisition cohort, we estimate

$$\Pr(60\text{-day delinq}_i = 1) = \alpha + \beta_1 \text{FICO}_i + \beta_2 \text{DTI}_i + \beta_3 \text{LTV}_i + \beta_4 \log \text{UPB}_i + \beta_5 \text{Small}_i + \varepsilon_i$$

as a linear probability model with HC3-robust standard errors. The small-loan indicator β_5 measures the residual delinquency premium for sub-\$150,000 loans after FICO/DTI/LTV controls, providing an outcome-based test of whether higher denial rates and rate premia for small loans are justified by realized credit risk.

5 Results

5.1 Baseline and Expanded Specifications

Table 1 presents the expanded-controls logit estimate for Truist 2021 to 2023. With LTV, loan type, occupancy, AUS, manufactured housing, conforming-loan flag, lien status, 30-year flag, and missingness indicators included, the Black-White adjusted approval odds ratio is 0.575 (95% CI [0.555, 0.595], $z = -31.35$, $p < 0.001$). The coefficient on Hispanic is 1.064 ($p < 0.001$), and on Asian, 0.690 ($p < 0.001$). The same expanded-controls model estimated with listwise deletion of records missing LTV, DTI, or income (rather than retaining those records via missingness indicators) yields a Black OR of 0.526, indicating that the headline estimate is not an artifact of how missingness is handled.

5.2 Conventional-Only Subsample

Excluding FHA, VA, USDA, and manufactured housing applications, which are known to receive distinctive underwriting treatment, yields a Black OR of 0.565 ($p < 0.001$) on a sample of 326,323 applications. The Hispanic and Asian coefficients are essentially unchanged from the full sample. The conventional-only finding rules out product-mix as a driver of the headline result.

Table 1: Expanded Logit: Truist Mortgage Approval, 2021 to 2023

Variable	OR	95% CI Lower	95% CI Upper	<i>p</i> -value
Log income	1.069	1.048	1.092	< 0.001
Log loan amount	1.573	1.544	1.604	< 0.001
DTI midpoint	0.902	0.901	0.903	< 0.001
LTV	0.985	0.985	0.986	< 0.001
Purpose: purchase	2.429	2.329	2.534	< 0.001
Purpose: refinance	1.634	1.579	1.691	< 0.001
Purpose: cash-out refi	1.110	1.053	1.170	< 0.001
FHA	0.970	0.890	1.057	0.485
VA	12.503	10.439	14.975	< 0.001
USDA	0.575	0.484	0.684	< 0.001
Investment property	0.597	0.558	0.639	< 0.001
Second home	0.435	0.411	0.460	< 0.001
Subordinate lien	2.270	2.190	2.352	< 0.001
Manufactured housing	0.416	0.379	0.457	< 0.001
30-year term	1.198	1.165	1.232	< 0.001
Conforming-limit	2.250	2.135	2.371	< 0.001
Missing LTV indicator	0.001	0.001	0.002	< 0.001
Missing DTI indicator	2.255	2.201	2.310	< 0.001
Missing income indicator	0.801	0.751	0.853	< 0.001
Black	0.575	0.555	0.595	< 0.001
Hispanic	1.064	1.031	1.098	< 0.001
Asian	0.690	0.659	0.723	< 0.001

Notes: N = 336,332. Logistic regression with action codes 1 (originated) and 3 (denied). White (non-Hispanic) is the reference racial category. AUS dummies were estimated but produced near-collinear coefficients in the full-sample logit; AUS effects are reported via stratified estimation in Section 5.7. McFadden pseudo- $R^2 = 0.498$.

5.3 Geographic Robustness

Table 2 reports geographic-FE robustness. Adding state fixed effects reduces the Black OR from 0.526 to 0.520, a movement of less than 1.2%. The within-county linear probability model, which absorbs all county-level unobserved heterogeneity by demeaning, produces a Black coefficient of -0.114 percentage points ($p < 0.001$), consistent in direction and statistical significance with the logit estimates. The robustness of the racial gap to county-level absorption is strong evidence that the disparity is not a compositional artifact of geographic concentration.

5.4 Denial Reason Decomposition

Table 3 reports the denial-reason decomposition using originated applications as the reference. The credit history coefficient is the largest in the table: Black applicants are 3.17 times more likely than comparable White applicants to receive a credit history denial relative to being originated.

Table 2: Geographic Fixed-Effects Robustness

Specification	Black OR / Coef	95% CI Lower	95% CI Upper	<i>p</i> -value
No geographic FE	0.526	0.511	0.541	< 0.001
State FE	0.520	0.505	0.535	< 0.001
County FE (LPM, pp)	−0.114	−0.119	−0.108	< 0.001

Notes: N = 336,332 in all specifications. The county-FE LPM uses within-county demeaning rather than a dummy-variable expansion to manage memory; the coefficient is interpreted as the percentage-point change in the probability of approval associated with the Black indicator, holding all other controls constant within county.

The DTI coefficient is 1.69, the incomplete-application coefficient is 1.40, and the other-denial coefficient is 1.66. The collateral coefficient, however, is 1.04 with a *p*-value of 0.363, statistically indistinguishable from parity.

Table 3: Denial Reason Decomposition: Binary Logit vs Originated

Denial Reason	N denied	Black OR	95% CI Lower	95% CI Upper	<i>p</i> -value
Credit history	40,421	3.174	3.065	3.286	< 0.001
DTI	40,670	1.693	1.595	1.797	< 0.001
Collateral	6,031	1.044	0.952	1.144	0.363
Incomplete application	23,298	1.398	1.331	1.468	< 0.001
Other denial	13,610	1.664	1.572	1.761	< 0.001

Notes: Each row reports the result of a separate binary logit estimated on the subsample of applications that were either originated (action code 1) or received the specific denial reason listed. Reference category in all rows is originated applications. Controls: log income, log loan amount, DTI midpoint, LTV, loan purpose dummies, FHA/VA flags, AUS dummies. White (non-Hispanic) is the racial reference. The collateral coefficient is the only coefficient in the table that is not statistically distinguishable from 1.0, suggesting appraisal bias is not a meaningful contributor to the racial disparity at this institution.

The 3.17 credit history coefficient is substantially larger than the 2.07 obtained in the simpler denial-conditional specification (the credit-history-versus-DTI model that conditions on having been denied). This is the expected direction of bias: the denial-conditional specification compresses the estimate by including only denied applicants, all of whom had some unfavorable signal that triggered review. The vs-originated specification more accurately captures the structural over-representation of credit history citations among Black applicants.

The collateral null is informative. If the racial disparity were primarily driven by appraisal bias, a mechanism that has received recent academic and regulatory attention, we would expect collateral denials to be disproportionately Black. They are not. The credit history channel, by contrast, is the dominant locus.

5.5 Loan-Size Gradient and Interaction

Table 4 reports Black ORs estimated separately within each loan-amount quartile of the Truist 2021 to 2023 sample, applying the full expanded-controls specification. The OR rises monotonically from 0.491 in the smallest quartile (median \$45,000) to 0.748 in the largest quartile (median \$435,000), a 51% relative widening. The estimated $\text{Black} \times \text{LoanQ1}$ interaction coefficient in a pooled specification is $\text{OR} = 0.807$ (95% CI [0.750, 0.869], $p < 0.001$), implying that the racial penalty in the smallest loan segment is 19% larger than the average racial penalty across all loan sizes.

Table 4: Black Approval OR by Loan Amount Quartile, Truist 2021 to 2023

Quartile	Median Loan Amount	N	Black OR	<i>p</i> -value
Q1 (smallest)	\$45,000	88,716	0.491	< 0.001
Q2	\$105,000	84,715	0.538	< 0.001
Q3	\$195,000	78,937	0.634	< 0.001
Q4 (largest)	\$435,000	83,964	0.748	< 0.001
Black $\times$ LoanQ1 interaction		—	336,332	0.807 < 0.001

Notes: Within-quartile estimates apply the expanded-controls specification of Table 1. The interaction coefficient is the OR on the $\text{is_black} \times \text{loan_q1}$ product term in a pooled specification that also includes the main effects.

5.6 Cross-Institution Variation

Table 5 reports institution-specific Black coefficients from a linear probability model with year fixed effects, estimated separately on each of the ten institutions in our 2018 to 2023 panel. The estimates range from -5.7 percentage points at Bank of America (the smallest gap) to -17.9 percentage points at PNC Bank (the largest). Truist’s gap of -15.5 pp falls in the middle of the distribution. The Pearson correlation between log institution size (measured by application count) and gap magnitude is 0.746, indicating that larger institutions exhibit smaller racial disparities. This is consistent with a standardization mechanism in which scale enables investment in automated underwriting infrastructure that compresses discretionary variation.

Table 5: Institution-Specific Black-White Approval Gaps, 2018 to 2023

Institution	N	Black Coefficient (pp)	95% CI	p-value
Bank of America	1,801,165	-0.057	[-0.060, -0.055]	< 0.001
JPMorgan Chase	1,370,838	-0.070	[-0.072, -0.067]	< 0.001
Wells Fargo	2,144,681	-0.081	[-0.083, -0.079]	< 0.001
Huntington Bank	566,070	-0.125	[-0.131, -0.120]	< 0.001
Citizens Bank	788,014	-0.136	[-0.141, -0.132]	< 0.001
Regions Bank	469,221	-0.150	[-0.153, -0.146]	< 0.001
Truist Bank	666,816	-0.155	[-0.159, -0.151]	< 0.001
U.S. Bank	1,318,892	-0.166	[-0.170, -0.163]	< 0.001
Fifth Third Bank	382,586	-0.167	[-0.174, -0.160]	< 0.001
PNC Bank	752,781	-0.179	[-0.183, -0.174]	< 0.001
Pooled (inst+year FE)	10,261,064	-0.121	[-0.122, -0.120]	< 0.001

Notes: Each within-institution estimate is from a linear probability model with year fixed effects, full credit-and-underwriting controls, and HC3-robust standard errors. The pooled specification adds institution fixed effects. Pearson correlation between log institution size and gap magnitude across the ten institutions: $r = 0.746$.

5.7 Automated Underwriting System Stratification

Table 6 reports Black ORs estimated within each AUS stratum at Truist 2021 to 2023. The 60,692 applications routed through Desktop Underwriter (Fannie Mae’s automated system) yielded a near-collinear estimation outcome that we omit; the 36,407 applications routed through Loan Prospector (Freddie Mac’s system) produce a Black OR of 0.701; and the 239,233 applications subject to manual or unspecified-AUS routing produce a Black OR of 0.524, a gap that is 25% larger than the LP-routed gap. The pattern is consistent with the standardization-reduces-disparity hypothesis: where automated rule-based systems govern the underwriting decision, racial disparities are compressed; where human discretion is dominant, disparities are wider.

Table 6: Black-White Approval OR by Automated Underwriting Stratum, Truist 2021 to 2023

AUS Stratum	N	Black OR	p-value
Loan Prospector (automated, Freddie)	36,407	0.701	< 0.001
Manual or unspecified	239,233	0.524	< 0.001

Notes: Estimates from the expanded-controls logit specification, restricted to each AUS stratum. The Desktop Underwriter stratum (N = 60,692) is omitted due to a near-collinear estimation outcome that produced uninformative confidence intervals. The 25% gap between LP and manual ORs supports the standardization-reduces-disparity interpretation.

5.8 Event Study: BB&T-SunTrust Merger

Table 7 reports dynamic event-study coefficients for the BB&T-SunTrust merger using a linear probability model with institution-clustered standard errors. The static triple-interaction estimate (Black $\times$ Truist $\times$ Post) is 0.007 percentage points ($p = 0.308$), not statistically distinguishable from zero. The dynamic coefficients reveal a more nuanced pattern. The 2019 coefficient (the merger-announcement year, with the merger announced in February 2019 and completed in December 2019) is statistically significant at $p = 0.003$, indicating a violation of strict pre-trend parallelism. The 2020 coefficient is not statistically distinguishable from zero ($p = 0.229$). Among post-merger years, 2021 shows a positive and significant gap-narrowing effect (+3.7 pp, $p = 0.008$), 2022 a marginally significant positive effect ($p = 0.051$), and 2023 a significant gap-widening effect (−2.5 pp, $p = 0.004$).

Table 7: Dynamic Event Study: Black $\times$ Truist $\times$ Year (Reference: 2018)

Year	Coefficient (pp)	SE	95% CI	p -value	Period
2019	0.021	0.007	[0.007, 0.035]	0.003	PRE
2020	0.014	0.012	[−0.009, 0.038]	0.229	PRE
2021	0.037	0.014	[0.010, 0.065]	0.008	POST
2022	0.027	0.014	[−0.000, 0.053]	0.051	POST
2023	−0.025	0.009	[−0.042, −0.008]	0.004	POST

Notes: Linear probability model with institution-clustered standard errors. Sample: 8,524,394 applications across Truist plus six control institutions, 2018 to 2023. Coefficients are interpreted as the percentage-point change in the Black-White approval gap at Truist relative to controls in year t , with 2018 as the reference year. The static Black $\times$ Truist $\times$ Post coefficient (not shown) is 0.007 pp ($p = 0.308$). The 2019 coefficient is partially attributable to merger announcement effects (BB&T-SunTrust merger was announced in February 2019 and completed in December 2019); we discuss this identification challenge in the text.

The honest interpretation of these results is that the merger conclusion is more nuanced than a simple narrowing-of-gap story. The 2019 pre-trend failure is consistent with an anticipation effect: institutions that know they will soon be part of a larger combined entity may begin adjusting their underwriting practices before the legal close. We note this as an identification challenge rather than a clean rejection of the merger hypothesis. The post-merger pattern itself is mixed, with two of three post-years showing positive coefficients and 2023 showing a gap-widening, suggesting that any short-term standardization gains may have eroded.

We refrain from a strong causal claim about the merger and instead interpret the static and dynamic coefficients as suggestive evidence of standardization effects, with the cross-institution and

AUS analyses providing more robust support for the standardization mechanism.

5.9 Loan-Size Pricing: LLPA Benchmark

Table 8 reports the rate-premium decomposition. The unconditional small-loan rate premium is 10.10 basis points (Model A: rate as a function of FICO, DTI, LTV, log UPB, and a small-loan indicator; $p < 0.001$). Adding the loan-by-loan LLPA estimate as a regressor (Model B) reduces the small-loan coefficient to 9.96 basis points. The LLPA accounts for only 0.14 basis points of the observed premium, equivalent to 1.4% of the total. The residual 9.96 basis points is statistically and economically large and is unexplained by the published GSE pricing schedule.

Table 8: Small-Loan Rate Premium Decomposition, Fannie Mae 2021Q1

Specification	Small-loan premium (bps)	p -value
Model A: no LLPA control	10.10	< 0.001
Model B: with loan-level LLPA	9.96	< 0.001
LLPA-explained portion	0.14 (1.4%)	—
Residual unexplained	9.96 (98.6%)	—

Notes: Sample: 5,056,556 30-year fixed conventional loans in the 2021Q1 Fannie Mae acquisition cohort. Outcome: original interest rate (basis points). LLPA computed loan-by-loan from the published Fannie Mae April 2022 LLPA Matrix, including the standard 25 basis point small-balance LLPA for loans below \$150,000. HC3-robust standard errors.

The result narrows the space of explanations for the small-loan rate premium. The premium is not predominantly a mechanical pass-through of GSE pricing rules. It must reflect either (a) per-loan fixed servicing costs amortized over a smaller principal balance, (b) lender-specific over-screening of small-balance borrowers, or (c) some combination. Disentangling (a) from (b) requires loan-level cost data not available in our public sample.

5.10 Outcome-Based Test: GSE Delinquency

Table 9 reports descriptive statistics by loan-amount quartile and the delinquency regression result for the Fannie Mae 2021Q1 cohort. The median FICO is essentially flat across quartiles (776 in Q1 to 779 in Q4, a 3-point spread). The mean DTI varies modestly (31.6 in Q1 to 33.8 in Q4). The mean LTV is 62.1 in Q1 versus 69.7 in Q4. The 60-day delinquency rate is 0.31% in Q1 versus 0.36% in Q4. The interquartile range of FICO, however, is 27% wider in Q1 (61 points) than in

Q4 (48 points), indicating substantially higher dispersion of credit-quality signals in the small-loan segment.

Table 9: Loan Characteristics and Delinquency by Loan Amount Quartile, Fannie Mae 2021Q1

Quartile	N (loan-mo)	Median UPB	Median FICO	FICO IQR	Mean DTI	60-day Delinq Rate
Q1	18,217,621	\$130,000	776	61	31.6	0.31%
Q2	18,305,783	\$213,000	777	56	32.7	0.30%
Q3	18,167,270	\$305,000	778	54	33.2	0.29%
Q4	18,056,333	\$468,000	779	48	33.8	0.36%

Notes: Sample: 72,747,007 loan-month observations from the 2021Q1 Fannie Mae acquisition cohort. Quartile cutoffs are based on the original UPB distribution. The 60-day delinquency rate is the cumulative probability that a loan was 60 or more days delinquent at any month during the observation window.

The delinquency regression (60-day delinquency as a function of FICO, DTI, LTV, log UPB, and a small-loan indicator) yields a small-loan coefficient of 0.066 percentage points ($p < 0.001$): the small-loan delinquency premium is positive and statistically significant after FICO/DTI/LTV controls. However, the magnitude is small relative to the 0.31% Q1 base rate, equivalent to a roughly 21% relative increase in delinquency probability while the racial approval gap widens by 51% relative across the same loan-size range.

The combined picture from Tables 8 and 9 is consistent with neither a pure over-screening nor a pure risk-pricing interpretation. Small loans face a 10.10 basis point rate premium not explained by LLPA, alongside a 0.066 percentage point delinquency premium that is small relative to the rate premium and to the approval gap. The wider FICO interquartile range in the small-loan segment provides indirect support for a variance-based mechanism in which thin-file applicants face stricter approval thresholds, though we cannot directly observe credit-file depth in the GSE acquisition data.

6 Wealth-Foregone Counterfactuals

To translate the statistical disparities of Section 5 into a policy-relevant magnitude, we conduct a counterfactual exercise: hold the expanded-controls covariate vector for each Black applicant fixed and set the Black indicator to zero, then compute the predicted approval probability under the actual model and under this counterfactual. The difference is the per-applicant excess denial probability attributable to the racial coefficient.

Applied to the 32,703 Black applicants in Truist’s 2021 to 2023 sample, the per-applicant excess denial probability averages 11.86 percentage points. Aggregated over the three-year period, this implies 3,878 excess Black denials, or 1,293 annually. We translate to wealth using two methods. Method A applies an aggregate appreciation rate of 48% (the approximate cumulative gain implied by the FHFA all-transactions index over a typical ten-to-thirteen year holding window for the Truist footprint) to a representative home value of \$239,000, yielding \$114,720 in foregone home equity per excess denial. Method B applies a full equity-build model with a 13-year holding period: home appreciation at 4.5% annually for 13 years (\$184,555), principal paid via a 30-year fixed-rate amortization schedule at 6.5% with 10% down (\$47,483), and the recovered \$23,900 down payment, yielding \$255,938 per household. The implied annual wealth-foregone range is \$148.3M (Method A) to \$330.8M (Method B).

Under a counterfactual policy in which credit history denials of Black applicants were equalized to White rates, the dominant locus identified in Section 5.4, the implied additional homeowners per year range from 829 (under a 50% conversion of equalized denials to originations) to 1,327 (under an 80% conversion), and the implied annual wealth gain ranges from \$95.1M to \$339.6M. We emphasize that these are upper-bound counterfactuals that abstract from general-equilibrium effects on housing prices and from selection of applicants into the application pool. The wealth parameters reflect publicly available housing-cost and amortization assumptions and should be interpreted as indicative orders of magnitude rather than precise welfare estimates.

7 Limitations and Unresolved Identification Challenges

Three identification challenges remain that we cannot address with public data and that we wish to flag explicitly.

Causal mediation. A formal interventional decomposition of the racial coefficient into direct (taste-based) and indirect (credit-score-mediated) channels requires credit score as a measured mediator. HMDA does not include credit scores. The FHFA National Mortgage Database (Federal Housing Finance Agency, 2024) includes binned credit score ranges and offers a path to this analysis, but access requires a formal application and Institutional Review Board approval. We identify this

as the natural next step.

Loan-level profitability parity. A true outcome-based discrimination test compares group-level realized profitability or internal rates of return at the loan level. This requires matched origination, performance, loss-given-default, prepayment, and servicing data with race-of-applicant identifiers. The Fannie Mae performance file we use for the delinquency analysis lacks loss-given-default and servicing fields and does not preserve race information from the HMDA application. Our delinquency analysis is therefore a partial substitute. Full profitability parity testing requires either internal lender data or a substantial enhancement to the GSE public-use files.

Credit file depth and thin-file heterogeneity. The variance-based mechanism predicts that minority applicants are disproportionately thin-file and therefore subject to stricter approval thresholds, which would generate both higher denial rates and pricing premia even at equal central tendencies of observable risk metrics. We document a 27% wider FICO interquartile range in the small-loan segment, which is consistent with this mechanism but does not directly measure trade-line counts or file age. Direct testing requires credit bureau data linked to HMDA records, which is not publicly available.

We do not view these limitations as undermining the core findings. The expanded-controls Black OR of 0.575, the credit history coefficient of 3.17, the AUS gap (OR 0.524 manual versus 0.701 LP), the cross-institution standardization correlation of $r = 0.746$, and the LLPA-residual rate premium of 9.96 basis points are all well-identified within our public-data sample. The three remaining gaps would tighten causal interpretation but would not, plausibly, reverse the direction or magnitude of the disparities documented here.

8 Conclusion

We document substantial racial disparities in U.S. mortgage approvals that survive an extensive set of underwriting controls available in the post-2018 HMDA expansion. The disparity is primarily located in credit history evaluation rather than in collateral assessment or DTI screening: Black applicants face an adjusted credit history denial relative to origination of $OR = 3.17$, while the collateral disparity is statistically indistinguishable from zero. The gap is monotonically larger in

smaller loans, and a formal interaction confirms a 19% amplification of the racial penalty in the smallest loan-amount quartile. Cross-institution variation is systematic and large, with a 0.746 correlation between log institution size and gap magnitude, supporting a standardization mechanism. Within institutions, the AUS stratification produces a 25% smaller racial gap under Loan Prospector-routed underwriting than under manual review, providing direct micro-level evidence for the same mechanism.

A loan-level benchmark against the published Fannie Mae LLPA schedule shows that the small-loan rate premium of 10.10 basis points is not explained by mechanical GSE pricing rules: only 0.14 basis points (1.4%) is attributable to LLPA, and the residual 9.96 basis points is unexplained. An outcome-based delinquency test using GSE performance data shows a small-loan delinquency premium of 0.066 percentage points, statistically significant but economically small relative to the rate premium and the approval gap, and inconsistent with a pure risk-pricing interpretation. Higher FICO interquartile-range dispersion in the small-loan segment (61 versus 48 points) provides indirect support for a variance-based mechanism.

The implied wealth-accumulation foregone, in the range of \$148M to \$331M annually for the Truist 2021 to 2023 sample alone, underscores the policy stakes. Three identification challenges remain: causal mediation requires credit scores not in HMDA, full profitability parity requires loss data not in the GSE public-use files, and direct testing of the thin-file mechanism requires credit bureau linkage. The FHFA National Mortgage Database is the natural next-step data source. In the meantime, the evidence assembled here supports two policy-relevant conclusions: standardized underwriting infrastructure compresses racial disparities, and credit history evaluation is the highest-priority locus for fair-lending audits at large depository institutions.

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